Cloud-Security-Assignment-3

Q1. Enable Multi-Factor Authentication (MFA) on your personal AWS or Azure cloud account using the built-in authenticator app option, and take a screenshot of the MFA setup confirmation page.

Ans:

Enable Multi-Factor Authentication (MFA) on an AWS Account

Step 1: Open Security Credentials

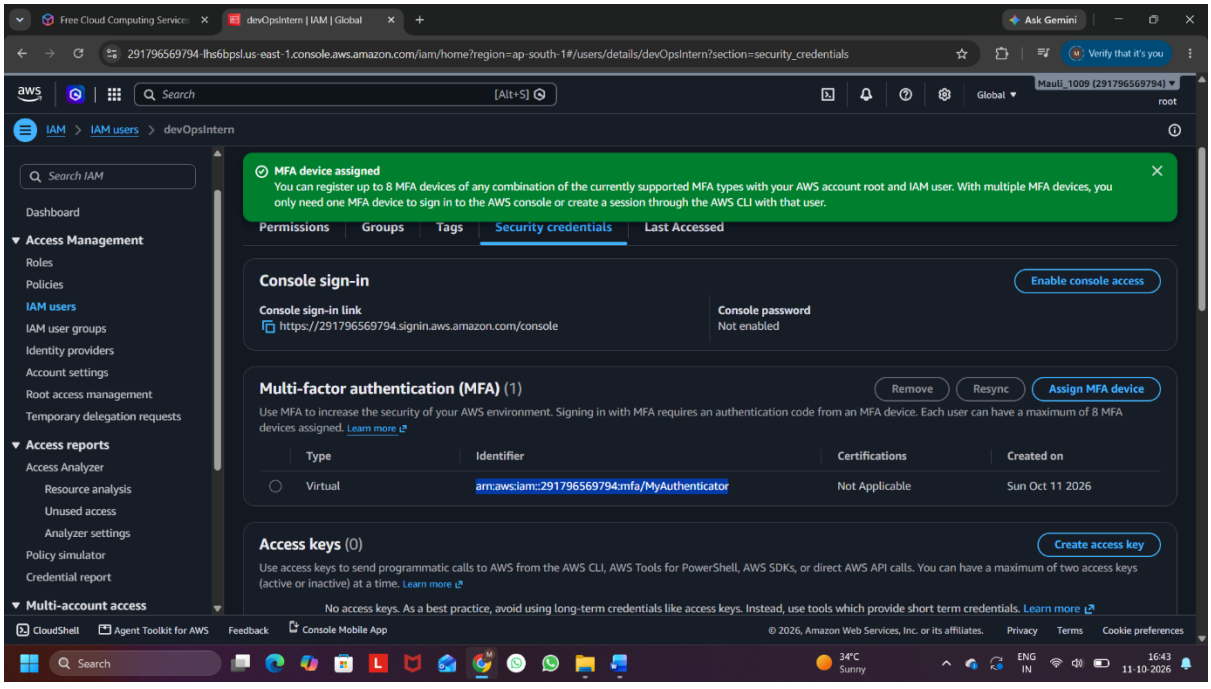
I logged in to the AWS Management Console, opened IAM, selected my user account, and navigated to the Security credentials section.

Step 2: Configure MFA

I selected the option to assign an MFA device and chose the authenticator app option. I scanned the displayed QR code using an authenticator app and entered the generated verification codes to complete the setup.

Step 3: Verify MFA Setup

After completing the configuration, I checked the MFA status to confirm that the authenticator device was enabled.



Screenshot 1: MFA Successfully Enabled Using an Authenticator App

Result: MFA was already enabled on the AWS Root account. For this practical, MFA was configured on an IAM user using the authenticator app, adding an extra layer of security to the user's sign-in process.

Q2. Simulate a login to your cloud account from a new browser or incognito mode and document the steps and prompts you receive with MFA enabled.

Ans:

Simulate a Login to AWS with MFA Enabled

Step 1: Open a New Browser Session

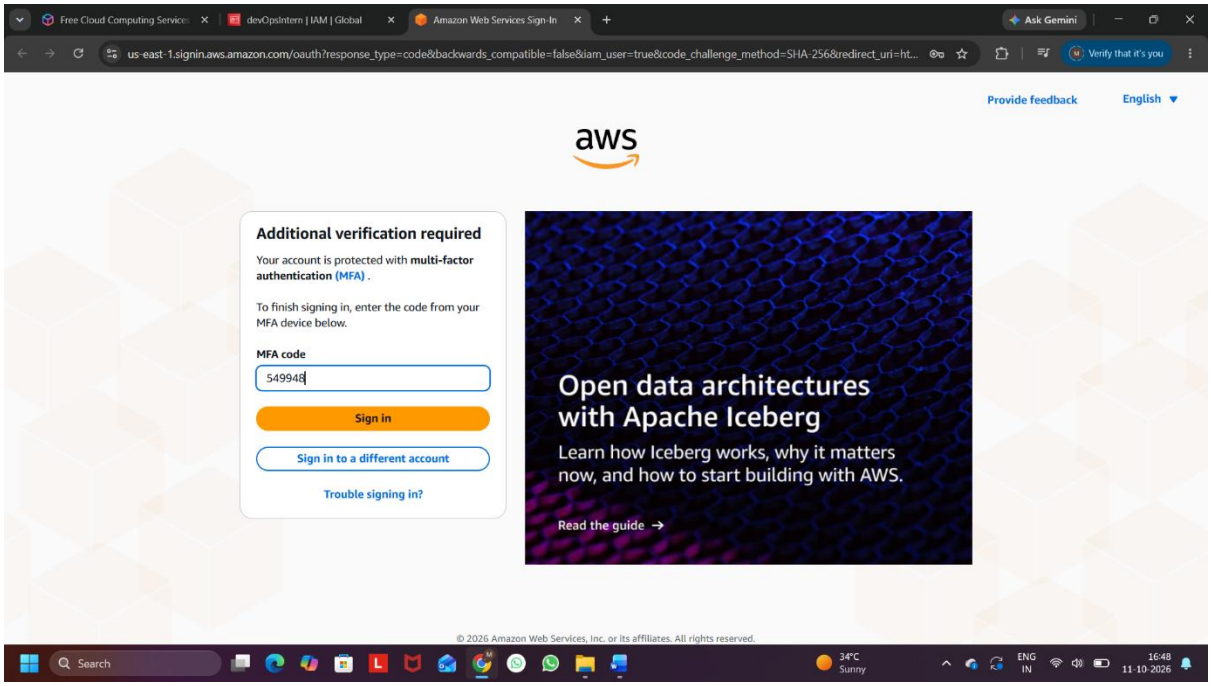
I opened a new browser session to simulate a fresh login to the AWS Management Console. Since MFA was already enabled on my AWS Root account, I used the IAM user configured with MFA to demonstrate the authentication process.

Step 2: Enter IAM User Credentials

I entered the IAM user's login credentials to access the AWS Management Console.

Step 3: Complete MFA Verification

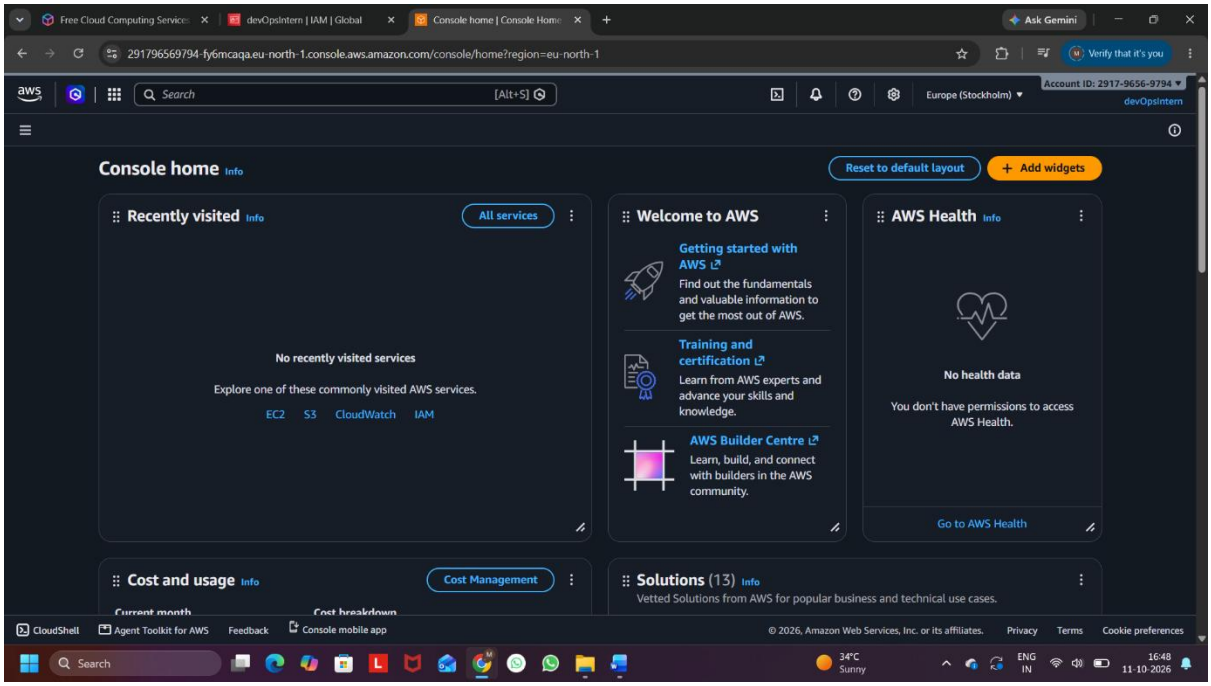
After entering the credentials, I followed the MFA verification prompt and entered the current code generated by the authenticator app associated with the IAM user.



Screenshot 1: MFA Verification Prompt for the IAM User

Step 4: Verify Successful Login

After successful MFA verification, the AWS Management Console opened, confirming that the IAM user's sign-in process required the additional authentication step.



Screenshot 2: AWS Console After Successful MFA Verification

Result: MFA was already enabled on the AWS Root account. For this practical, an IAM user with MFA enabled was used to demonstrate the login verification process in a new browser session. This provides an additional layer of security by requiring an authenticator-generated code along with the user's login credentials.

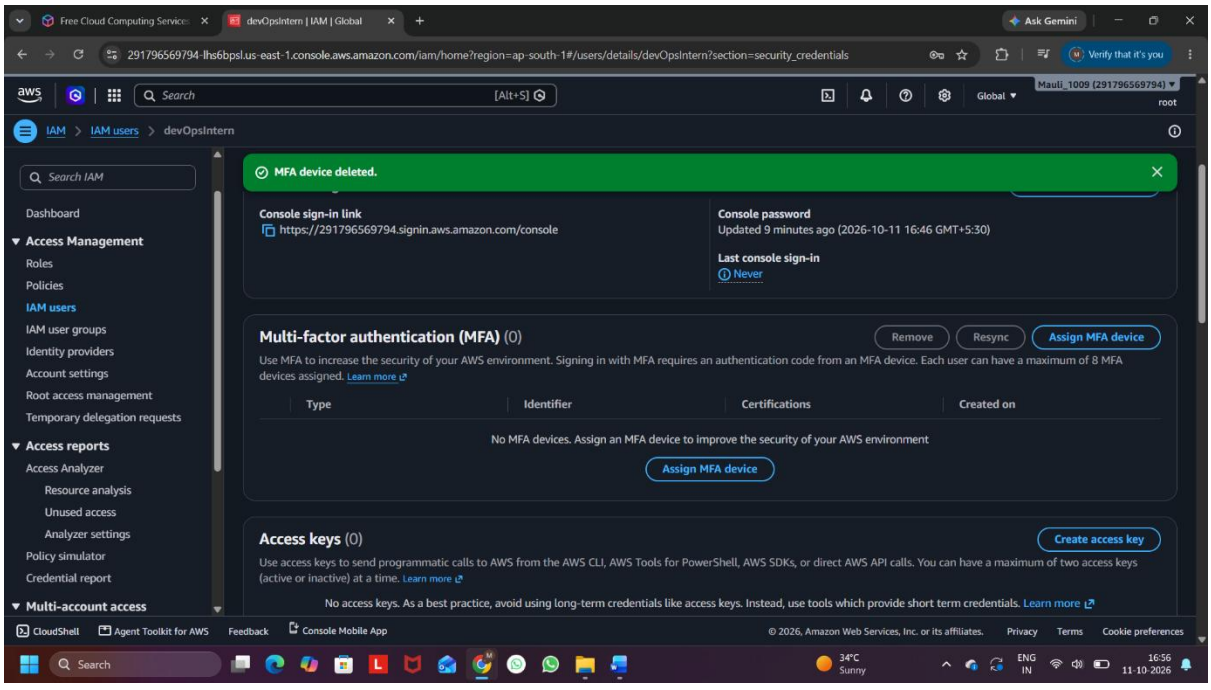
Q3. Disable MFA temporarily on your cloud account, attempt to log in again, and note the difference in the login experience and security prompts. Hint: Compare the number of steps and any warnings shown with and without MFA.

Ans:

Compare AWS Login With and Without MFA

Step 1: Temporarily Disable MFA

I temporarily disabled MFA on the IAM user used for the practical while keeping MFA enabled on the AWS Root account.



Screenshot 1: IAM User MFA Status After Deactivation

Step 2: Attempt to Log In Again

I opened a new browser session and entered the IAM user's username and password. I observed whether AWS requested an additional MFA verification

code.
I can access AWS without any MFA code and only using username and password,

Step 3: Compare the Login Experience

With MFA enabled, login required the user's credentials and an additional authenticator-generated verification code. With MFA disabled, the login process generally required only the username and password, without the extra MFA verification step.

Result: Disabling MFA reduces the number of authentication steps but also reduces account security. MFA should remain enabled to provide an additional layer of protection against unauthorized access.

Q4. Write a short scenario (3-5 lines) describing how an attacker might try to access a cloud account without MFA, and explain how enabling MFA would block or alert you to this attempt.

Ans:

An attacker obtains a user's AWS password through a phishing email and tries to log in to the cloud account. Without MFA, the attacker may gain access using only the stolen password. With MFA enabled, the attacker must also provide a verification code from the user's authenticator app. Without this code, the login attempt is blocked, providing an additional layer of security. Unexpected login attempts may also trigger security alerts if monitoring is configured.